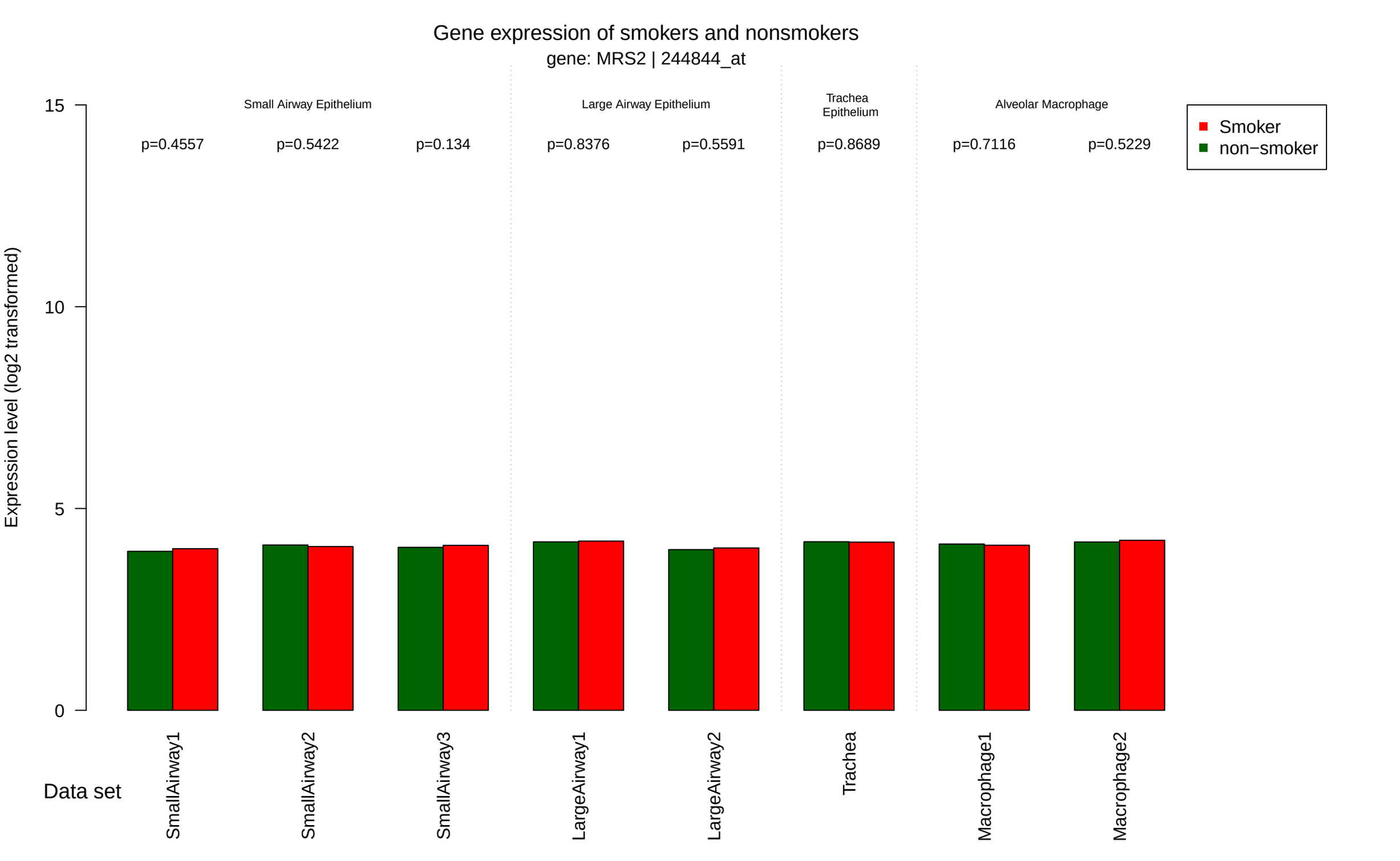
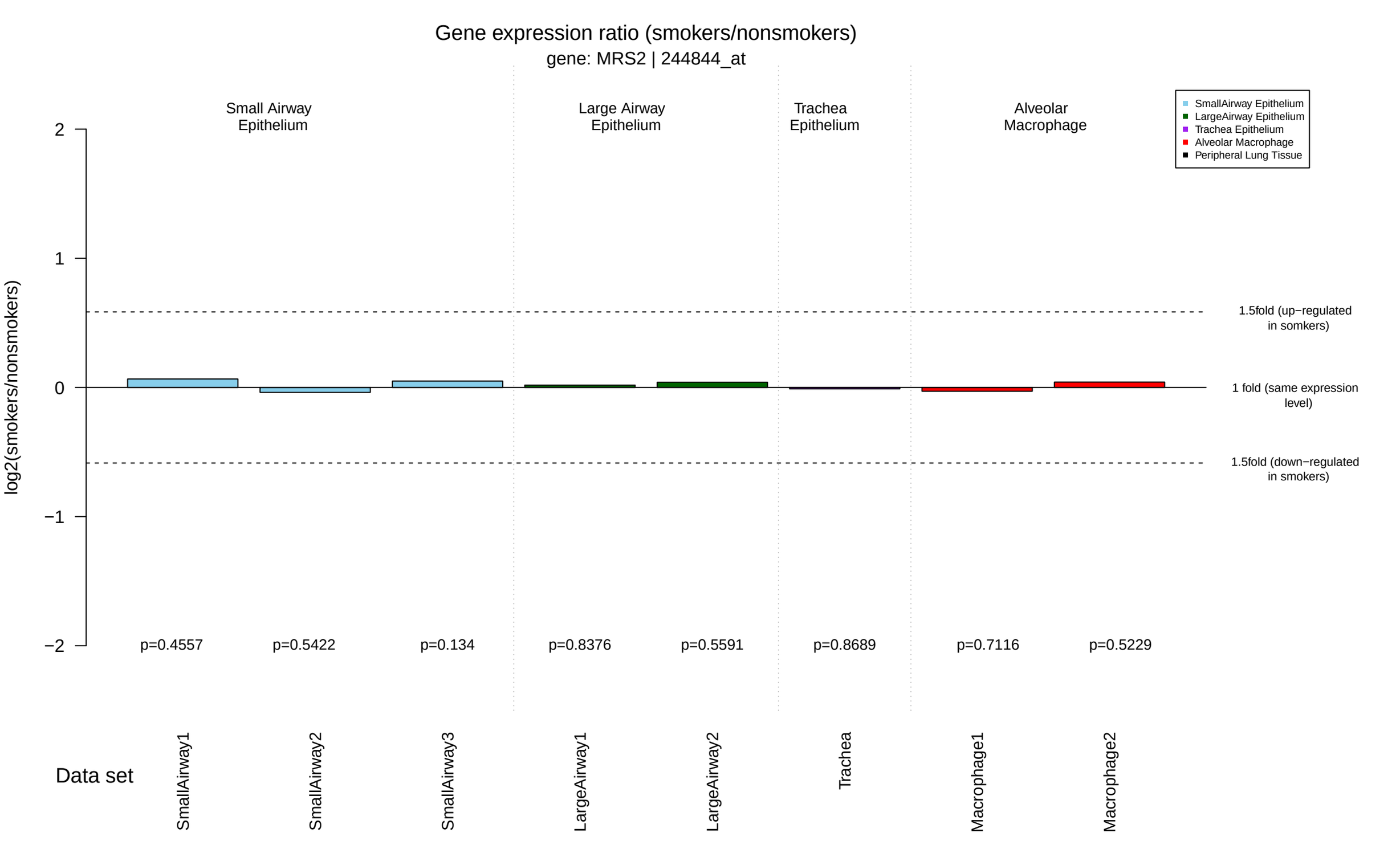
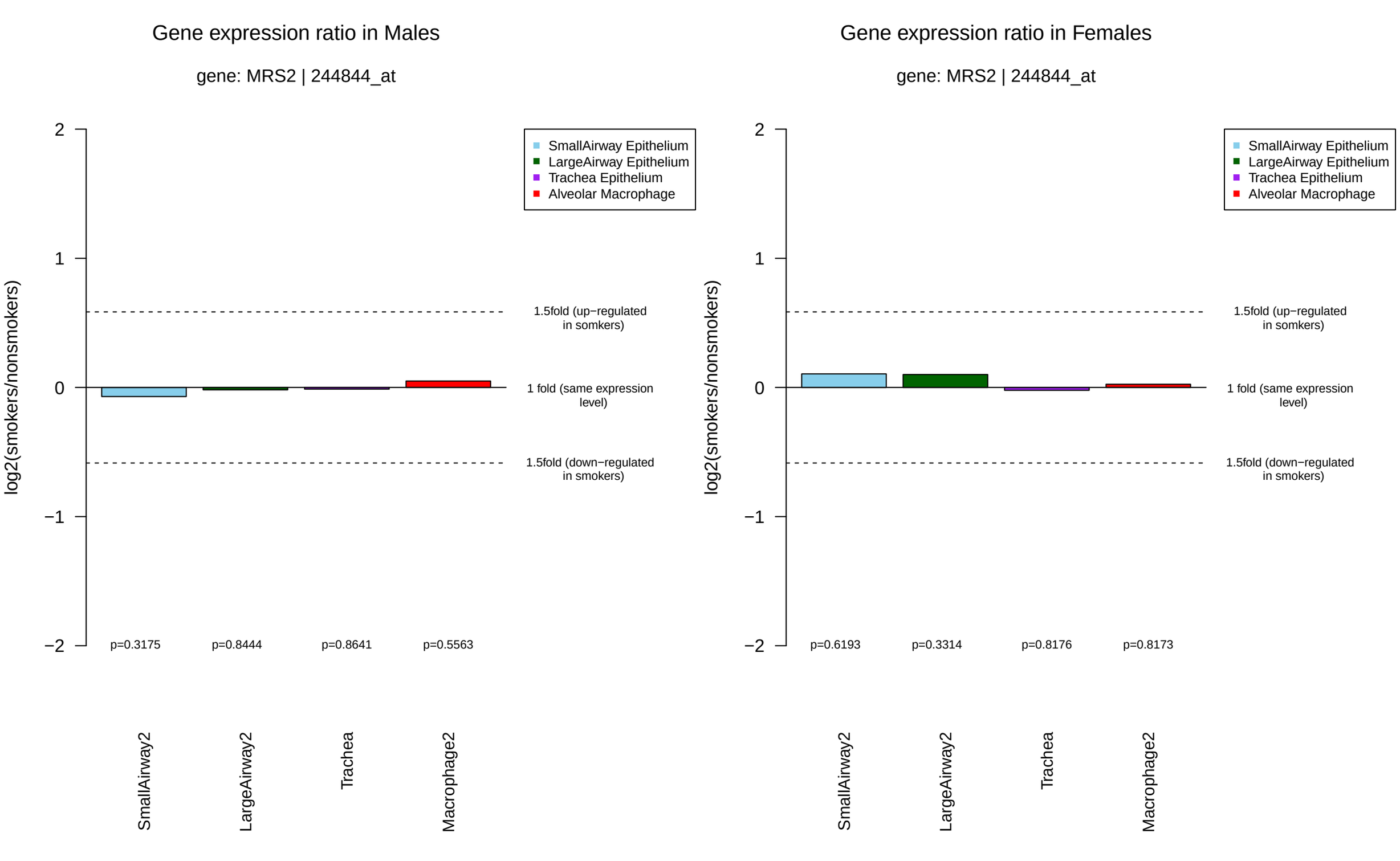
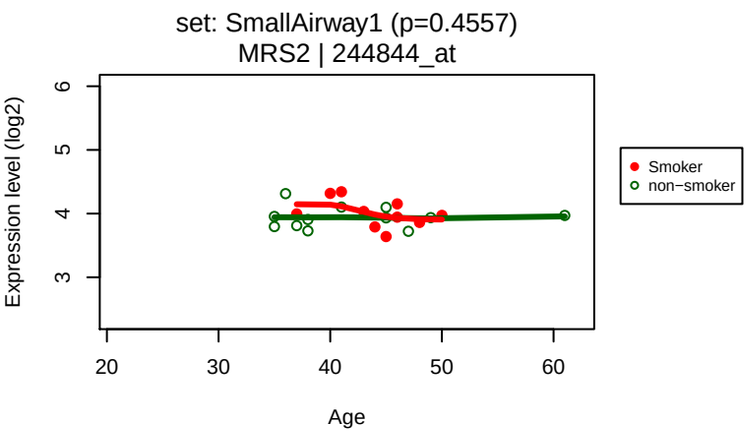
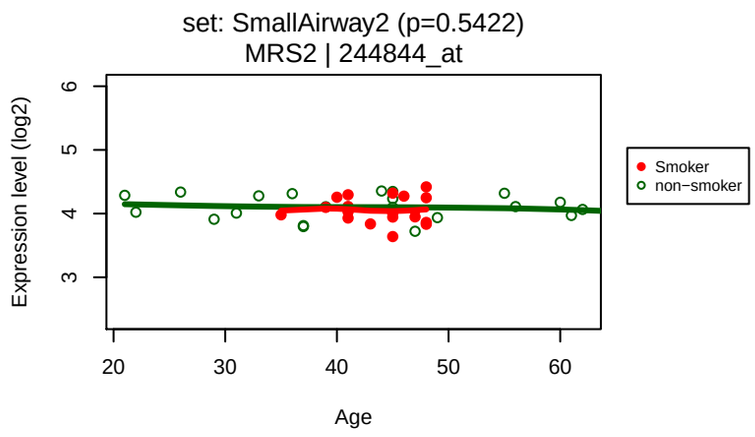
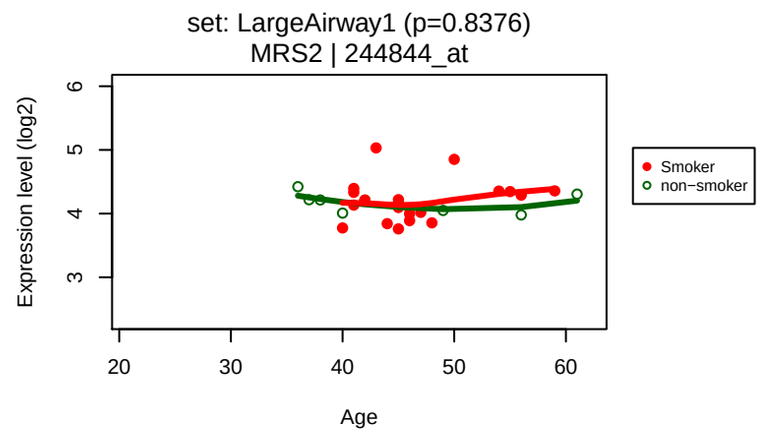

Gene expression of smokers and nonsmokers
gene: MRS2 | 244844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4557
p=0.5422
p=0.134
p=0.8376
p=0.5591
p=0.8689
p=0.7116
p=0.5229
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

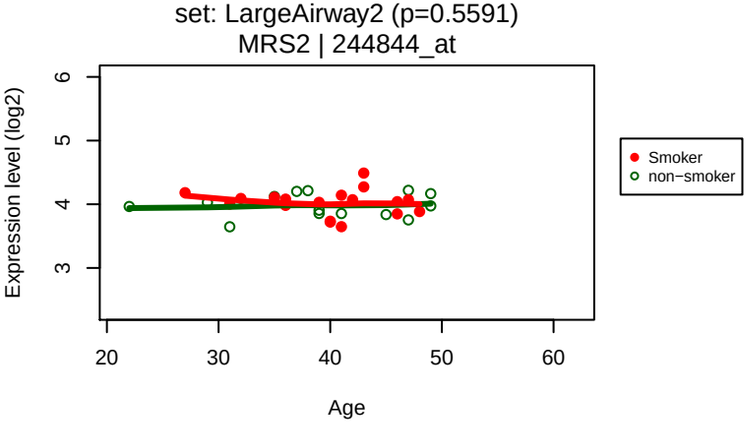

Gene expression ratio (smokers/nonsmokers)
gene: MRS2 | 244844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4557
p=0.5422
p=0.134
p=0.8376
p=0.5591
p=0.8689
p=0.7116
p=0.5229
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

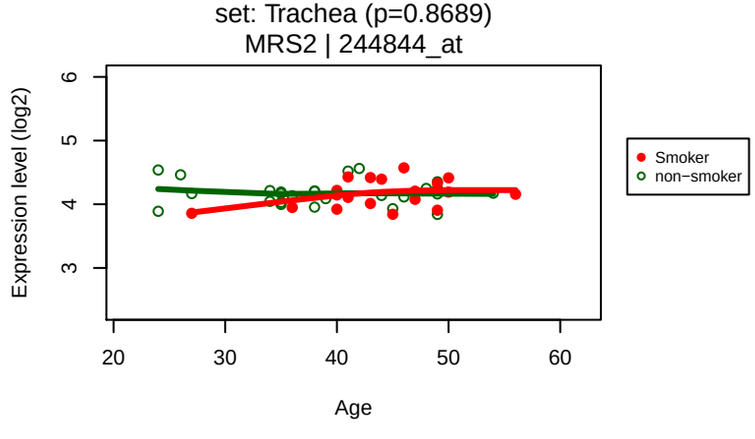

Gene expression ratio in Males
gene: MRS2 | 244844_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3175
p=0.8444
p=0.8641
p=0.5563
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRS2 | 244844_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6193
p=0.3314
p=0.8176
p=0.8173
SmallAirway2
LargeAirway2
Trachea
Macrophage2

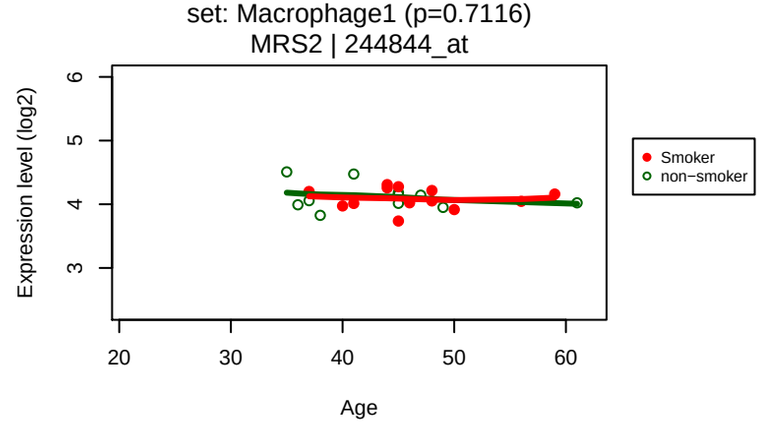

set: SmallAirway1 (p=0.4557)
MRS2 | 244844_at
Expression level (log2)
Age
Smoker
non−smoker

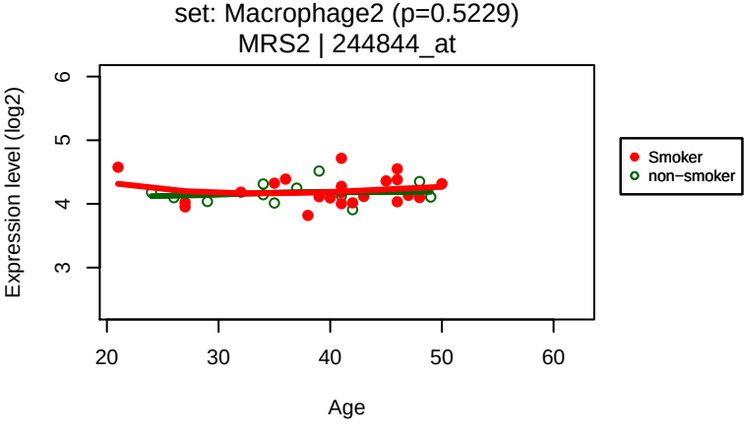

set: SmallAirway2 (p=0.5422)
MRS2 | 244844_at
Expression level (log2)
Age
Smoker
non−smoker

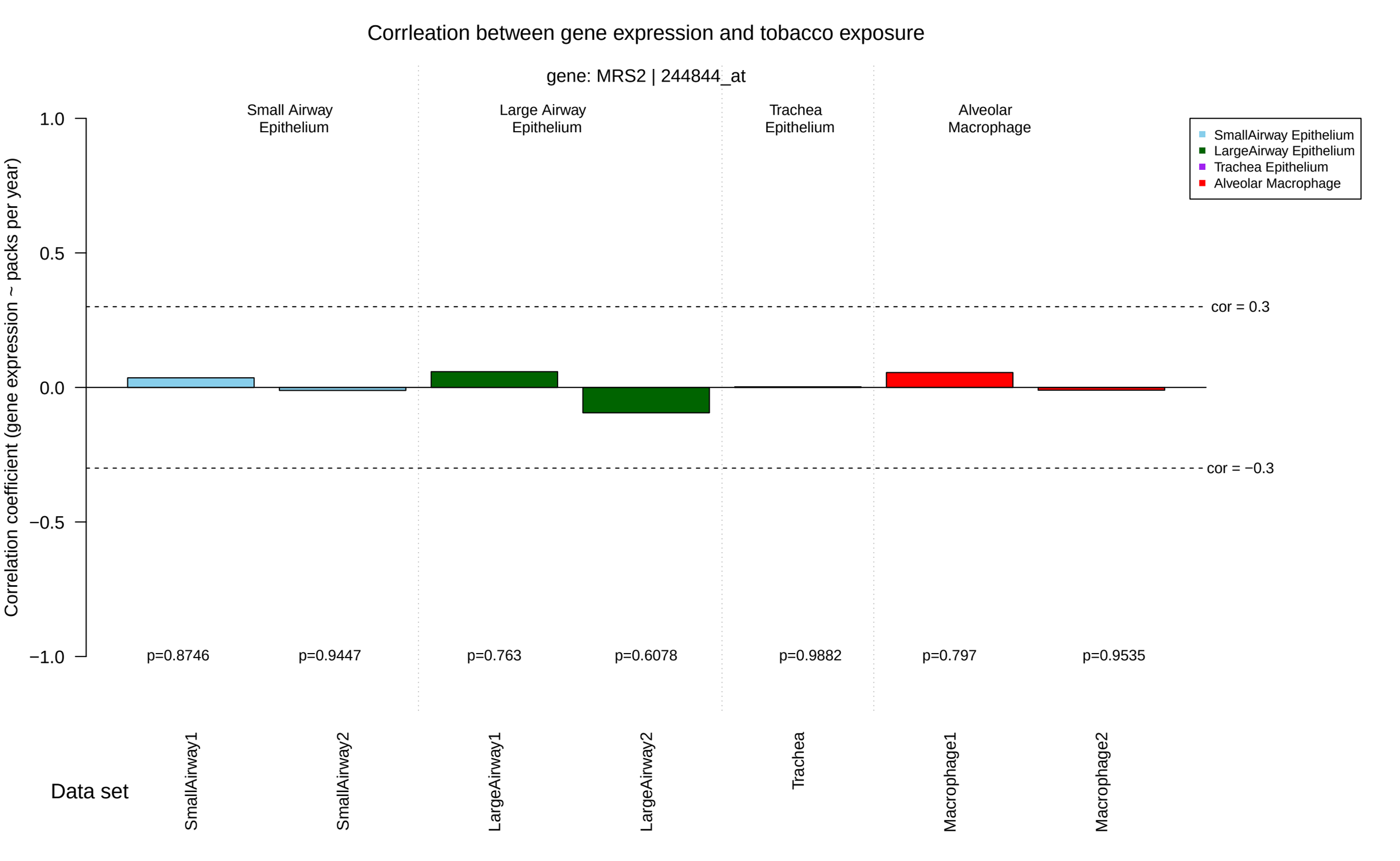

set: LargeAirway1 (p=0.8376)
MRS2 | 244844_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5591)
MRS2 | 244844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8689)
MRS2 | 244844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7116)
MRS2 | 244844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5229)
MRS2 | 244844_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRS2 | 244844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8746
p=0.9447
p=0.763
p=0.6078
p=0.9882
p=0.797
p=0.9535
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2